

Day 5: Network Testing Tools Theory

Task 1 :- Use the ping command to test connectivity from your computer to www.youtube.com and note the average response time.

ANS :-

At the bottom of the output, look for Average = XXms (or round-trip min/avg/max on Mac/Linux) :

- Under 20 ms: Excellent response time (ultra-fast).
- 20 – 50 ms: Good, smooth connection (standard for most broadband connections).
- 50 – 100 ms: Average, acceptable for video streaming without buffering.
- Over 100 ms: Slow response time (may cause delayed loading or buffering).

Here is screenshot :-

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Microsoft Windows [Version 10.0.26200.9278]
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C:\Users\asvin>ping www.youtube.com

Pinging www.youtube.com [2001:4860:482b:400::] with 32 bytes of data:
Reply from 2001:4860:482b:400::: time=26ms
Reply from 2001:4860:482b:400::: time=30ms
Reply from 2001:4860:482b:400::: time=37ms
Reply from 2001:4860:482b:400::: time=51ms

Ping statistics for 2001:4860:482b:400:::
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 26ms, Maximum = 51ms, Average = 36ms

C:\Users\asvin>
```

Task 2 :- Run the traceroute (tracert on Windows) command to www.flipkart.com and list all the intermediate hops shown in the output.

ANS :-

Each numbered line in your output represents an intermediate hop (a router/node your request passes through):

- Hop 1: Your home router (usually 192.168.1.1 or 192.168.0.1).
- Hops 2–5: Your Internet Service Provider (ISP's local network and servers).
- Middle Hops: Major internet backbone routers transferring data across regions or cities.
- Final Hop: Flipkart's destination server or content delivery network (CDN).

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Microsoft Windows [Version 10.0.26200.9278]
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C:\Users\asvin>tracert www.flipkart.com

Tracing route to flipkart.com [64:ff9b::67f3:205a]
over a maximum of 30 hops:

  1    2 ms    2 ms    1 ms  2409:40c1:4107:1604::55
  2    *      *      *      Request timed out.
  3   69 ms   17 ms   20 ms  2405:200:5210:3:3925::ff02
  4   64 ms   35 ms   19 ms  2405:200:898:3632:62::6
  5   12 ms   13 ms   10 ms  64:ff9b::c0a8:e3c2
  6   31 ms   18 ms   18 ms  64:ff9b::c0a8:f48a
  7    *      *      *      Request timed out.
  8    *      |
```

Task 3 :- Use the nslookup tool to find the IP address of www.zomato.com and record the result.

ANS :-

```
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C:\Users\asvin>nslookup www.zomato.com
Server: UnKnown
Address: 10.86.106.3

Non-authoritative answer:
Name: e70929.dsca.akamaiedge.net
Addresses: 2600:140f:4::17da:f99a
           2600:140f:4::17da:f9b3
           184.86.248.194
           184.86.248.161
Aliases: www.zomato.com
          zomato.edgekey.net

C:\Users\asvin>
```

Task 4 :- Test connectivity to another device on your local network (such as your friend's laptop or your phone) using the ping command and describe what happens if the device is offline.

ANS :-

What Happens When the Device is Offline :-

If the target device is turned off, disconnected from Wi-Fi, or blocking ICMP ping requests with a firewall, you will see one of these main responses :

- Request timed out :-
 - Means : Your computer sent out data packets waiting for a response, but the offline device never replied within the time limit.

- Destination host unreachable :-
 - Means : Your router or computer doesn't know where to route the packet because the target IP address isn't active on the local network (no active MAC address association).

- 100% Packet Loss :-
 - At the end of the summary, it will state: Packets: Sent = 4, Received = 0, Lost = 4 (100% loss).

```
Microsoft Windows [Version 10.0.26200.9278]
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C:\Users\asvin>ping 192.168.2.5

Pinging 192.168.2.5 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.2.5:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Users\asvin>
```

Task 5 :- Create a short report (5-7 lines) summarizing which tool (ping, traceroute, nslookup) is most useful for diagnosing each of these problems: (a) Website not loading, (b) Slow connection, (c) Suspecting DNS issues.

ANS :-

- **(a) Website not loading :**

Use ping. It quickly checks if the web server is online or if your device has basic internet connectivity.

- **(b) Slow connection :**

Use traceroute (or tracert). It maps the entire path of your connection and pinpoints exactly which router or node is causing delays or dropping data.

- **(c) Suspecting DNS issues :**

Use nslookup. It directly tests if your DNS server can translate a domain name (like google.com) into its corresponding IP address.